PROJECT REPORT: CALIFORNIA HOUSE PRICE PREDICTOR

Course/Task: Task 1 – Machine Learning Workflow Implementation

Organization: Internship Portfolio Submission for Maincrafts Technology

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1. Executive Summary & Objectives

The objective of this project is to implement an end-to-end Machine Learning pipeline to predict neighborhood housing values using the California Housing dataset. This project covers the essential stages of the machine learning lifecycle: data ingestion, exploratory data analysis (EDA), data preprocessing, model training, evaluation, and visualization. A Linear Regression baseline model was successfully deployed and monitored to understand core predictive structural trends.

Core Technology Stack:

- **Programming Language:** Python
- **Data Ingestion & Manipulation:** Pandas, NumPy
- **Modeling & Metrics:** Scikit-Learn (LinearRegression)
- **Data Visualization:** Matplotlib, Seaborn

2. Exploratory Data Analysis (EDA) & Insights

The California Housing dataset consists of 20,640 neighborhood block groups, tracking 8 independent numerical features alongside a continuous target variable (MedHouseVal — Median House Value).

2.1 Key Findings:

1. **Data Integrity:** Initial profiling confirmed that the dataset contains **zero missing or null values**, eliminating the need for data imputation strategies.
2. **Feature Correlation Analysis:** By generating a Pearson correlation matrix heatmap, the linear relationship between features was evaluated.
    - **The Dominant Predictor:** The feature MedInc (Median Income of block residents) demonstrates a strong positive correlation of approximately **0.69** relative to the target house value (MedHouseVal). This establishes neighborhood wealth as the single most critical economic indicator of real estate pricing within this dataset.
    - **Other Variables:** Features like total rooms (AveRooms) showed a mild positive correlation, whereas geographic variables (Latitude and Longitude) displayed complex spatial patterns.

3. Model Implementation & Partitioning

To ensure an unbiased evaluation of the model's generalized predictive capabilities, a strict train-test split strategy was adopted:

- **Training Partition:** 80% of the dataset was utilized to let the Linear Regression algorithm learn coefficients and intercepts.
- **Testing Partition:** 20% of the dataset was entirely sequestered to act as an unseen "final exam" to measure performance metrics.

A standard OLS (Ordinary Least Squares) Linear Regression model was fit to the training partitions:

$$\hat{y} = \theta_0 + \theta_1 x_1 + \theta_2 x_2 + ... + \theta_n x_n$$

4. Performance Evaluation & Scorecard

The trained baseline model was used to predict values on the unseen test set, yielding the following results:

- **Mean Absolute Error (MAE): 0.533**
  - *Interpretation:* On average, the model's price predictions deviate from actual property values by approximately **$53,300** (as the target is scaled in units of $100,000).
- **Root Mean Squared Error (RMSE): 0.746**
  - *Interpretation:* RMSE penalizes larger scale errors more heavily. Because the RMSE is notably higher than the MAE, it reveals that the model struggles and makes larger-than-average errors on premium, high-value outlier properties.
- **Coefficient of Determination (R^2 Score): 0.576**
  - *Interpretation:* The Linear Regression model successfully captures and explains **57.6%** of the total variance present in California housing prices.

4.1 Visual Error Inspection:

The generated "Actual vs. Predicted" scatter plot confirmed a clear upward diagonal trend, matching the line of perfect fit. However, a distinct horizontal compression boundary was observed at the 5.0 ($500,000) mark due to artificial data capping in the source dataset.

5. Analytical Limitations & Future Enhancements

While a baseline score of ~57.6% provides a stable benchmark, a simple linear model introduces specific constraints:

5.1 Limitations:

- **Linearity Constraint:** Linear Regression assumes a strictly straight relationship between features. Real estate economics rely on complex non-linear interactions (e.g., location value is non-linear based on proximity to coastlines or cities).
- **Data Saturation:** The target data capping at 5.0 introduces skewed errors at the upper margins of real estate pricing.

5.2 Proposed Enhancements:

1. **Feature Engineering & Scaling:** Applying a StandardScaler to handle variances in feature ranges (e.g., comparing raw populations to median income values).
2. **Advanced Algorithms:** Transitioning from linear baselines to non-linear, ensemble-based decision tree frameworks—such as **Random Forest Regressors** or **XGBoost (Gradient Boosting)**—to map non-linear interactions and push the target R^2 accuracy baseline above 80%.